



CYBERROVER X4.3

MODULAR DUAL-LAYER AUTONOMOUS EXPLORATION & ENVIRONMENTAL SCOUTING UGV

1ST PRIZE WINNER — STATE LEVEL SCIENCE EXHIBITION | RAMGARH 2026

• MOVE • SENSE • INSPECT • LOG

ENGINEERING PROJECT • 2026

WHEN THE ENVIRONMENT IS LETHAL, DON'T SEND THE HUMAN FIRST.

Human entry into coal mine collapses, chemical leaks, and fire voids carries severe asphyxiation and explosion risks. CyberRover X4.3 enters first to establish safety benchmarks and atmospheric hazard profiles before rescue operations commence.

OPERATIONAL FLOW

A four-stage proxy loop that keeps the human safe while delivering critical intelligence.



① HAZARD INGRESS

UGV deployed into collapsed or contaminated target zones



② ATMOSPHERIC SCAN

Multi-gas and microclimate sensors sample ambient air



③ LONG-RANGE BROADCAST

Sub-GHz LoRa penetrates rubble to transmit telemetry



④ COMMAND DECISION

Remote operators make informed rescue decisions from safe standoff distance

ONE PLATFORM. MULTIPLE SENSES.

The CyberRover X4.3 unifies five segregated, modular subsystems into an autonomous reconnaissance tool:

MOVE

4WD differential skid-steering drivetrain, Dual BTS7960 43A high-current H-bridges, High-clearance all-terrain treaded wheels

SENSE

3-Gas Array: Methane CH₄, Carbon Monoxide CO, Air Quality AQI; Climate: Dual Temp averaging, Humidity, Dew Point, Barometric Pressure

LOCATE

u-blox NEO-6M GPS 3D positioning, Real-time Latitude, Longitude, Satellites & Speed

BROADCAST

Reyax RYLR998 868/915 MHz sub-GHz link, 1+ km wireless range through rock and concrete

LOG

Laptop Ground Station Cockpit with live oscilloscope, Persistent SQLite mission database with 1-click CSV export

SEGREGATED DUAL-TIER HIERARCHY: ZERO MOTOR NOISE, ZERO BROWNOUTS

High-draw inductive motor noise is physically isolated from sensitive environmental radio instrumentation.

CONTROL PATH

Handheld Remote (ESP32) → ESP-NOW
Protocol → ESP32-S3 Master → Arduino
Uno → BTS7960 Drivers → 4x Geared DC
Motors

ENVIRONMENTAL PATH

MQ Gas Array + BMP280 + DHT11 + NEO-
6M GPS → Arduino Nano (Layer 2 Master)
→ 1.3" I2C OLED HUD

COLLISION RADAR PATH

3x Ultrasonic Sensors (Left / Center / Right)
→ Arduino Uno → Automated Emergency
Braking

TELEMETRY & DATABASE PATH

Arduino Nano → RYLR998 LoRa TX →
Laptop USB-TTL RX → Python Telemetry
Engine → SQLite DBMS & Mission Cockpit

BUILT FOR ROUGH VOIDS, DEBRIS, AND COALFIELD TERRAIN

The physical design prioritizes stability, distributed electronics, practical sensor placement, and sufficient ground clearance for prototype-stage operation in varied environments.

Chassis & Suspension

Layered vibration-isolated deck with high-traction rubber-treaded wheels.

Front Ultrasonic Shield

Triple HC-SR04 sonar array mounted in curved front bumper.

Rear Atmospheric Binnacle

MQ-4, MQ-7, and MQ-135 sensors positioned away from motor heat.

Rooftop Eye HUD Pod

Angled canopy housing 1.3" OLED running expressive procedural RoboEyes.

RF & Navigation Mast

Integrated GPS ceramic patch antenna and sub-GHz coiled LoRa whip antenna.

Forward Camera Mount

Universal adjustable phone bracket for optical reconnaissance.

4WD DIFFERENTIAL MOBILITY

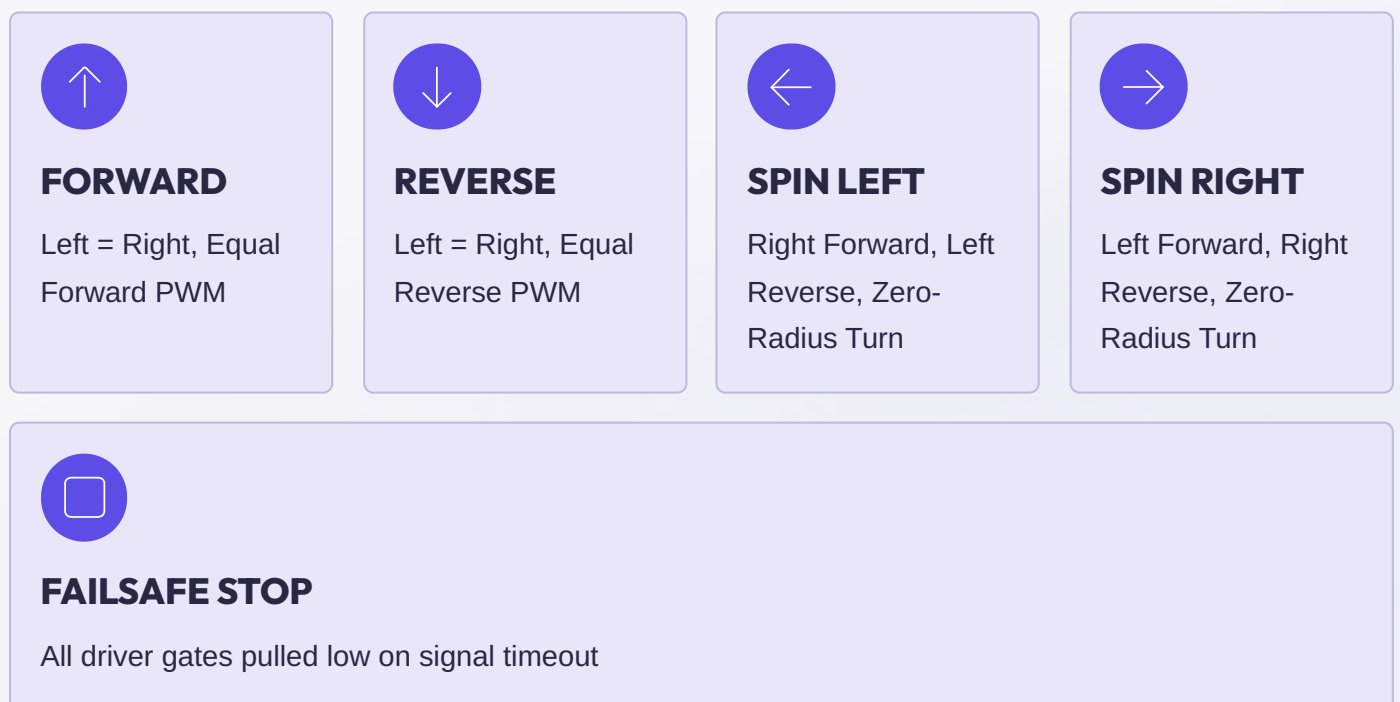
High-Torque Skid-Steering for Tight, Confined Voids

Four independent geared DC motors driven by dual 43A BTS7960 H-bridges provide pivot-in-place agility without mechanical steering links.

MOTOR CONTROL CHAIN



SKID-STEERING STATES



Skid-steering: Equal PWM → straight. Differential PWM between left and right groups → turning. No mechanical steering linkage required.

TRIPLE-RADAR COLLISION AVOIDANCE

Three Directions. One Real-Time Autonomous Decision.

Three ultrasonic transducers continuously sweep the forward hemisphere to prevent the rover from driving into fissures or fallen debris.

SENSOR POSITIONS

LEFT RADAR (45°)

Detects tunnel walls and corner obstacles

CENTER RADAR (0°)

Primary forward clearance and drop-off detector

RIGHT RADAR (45°)

Detects right-side obstructions

 Sensors trigger and read **individually in sequence** — not simultaneously.

SEQUENTIAL DETECTION CYCLE



1) TRIGGER LEFT → SAMPLE RETURN TIME → STORE DISTANCE



2) TRIGGER CENTER → SAMPLE RETURN TIME → STORE DISTANCE



3) TRIGGER RIGHT → SAMPLE RETURN TIME → STORE DISTANCE



4) THRESHOLD EVALUATION → IF DISTANCE < 25 CM, TRIGGER EMERGENCY AUTO-STOP BEFORE COMMAND OVERRIDE

ENVIRONMENTAL INSTRUMENTATION DECK

Atmospheric Hazard Detection & Power Calibration

The CyberRover X4.3 carries a comprehensive sensor suite into locations where direct human presence may be unsafe. Raw sensor data is collected by the Arduino Nano, transmitted via UART to the RYLR998 LoRa module, and broadcast to the ground station for environmental assessment.

SENSOR MODULES

MQ-4 (Pin A0) — Methane & Combustible Gas

Safety Threshold: 700 ADC. Methane and combustible gas indication for pre-entry hazard assessment.

MQ-7 (Pin A1) — Carbon Monoxide Poisonous Gas

Safety Threshold: 850 ADC. CO presence indication for fire-affected areas, enclosed spaces, and post-combustion zones.

MQ-135 (Pin A2) — Toxic Air Quality

Ammonia, VOCs, and broad-spectrum toxic air quality screening. Safety Threshold: 700 ADC.

BMP280 + DHT11 — Ambient Conditions

Averaged ambient temperature (°C), humidity (%), barometric pressure (hPa), and relative altitude (m).

u-blox NEO-6M GPS

3D fix, satellites, and precision latitude & longitude coordinates.

3S Li-ion Power Diagnostics

Calibrated 0–25V divider network with exact 0.852f trim calibration factor. Live 12.1V pack readout with graphical 9.6V (0%) to 12.6V (100%) fill gauge on OLED and Ground Cockpit.



Gas sensors output raw ADC values only. No certified ppm concentrations are calculated. Readings support **preliminary** environmental assessment — not definitive hazard classification.

SUB-GHz LoRa TELEMETRY PROTOCOL

Reliable Long-Range Packet Pipeline to Ground Station

Sensor data is packed into a compact CSV string broadcast every 1.5 seconds at 115200 baud via RYLR998 LoRa. The ground station receives, parses, and logs all telemetry into a persistent SQLite database.

TELEMETRY PACKET STRUCTURE (CR43 Envelope)

PACKET FORMAT

```
CR43,<pkt>,<volt>,<mq4>,<mq7>,<mq135>,<temp>,<hum>,<pres>,<alt>,<fix>,<lat>,<lon>,<sats>,<spd>
```

GROUND STATION PROCESSING FLOW

01

1) RADIO RECEPTION

Laptop USB-TTL adapter captures +RCV serial packet.

02

2) SIGNAL PROFILING

RSSI (dBm) and SNR (dB) recorded for link budget verification.

03

3) SQLITE STORAGE

Telemetry row committed with ISO-8601 millisecond timestamps.

04

4) LIVE COCKPIT UPDATE

Gauges, oscilloscope curves, and 3D artificial horizon update immediately.

OBSERVE

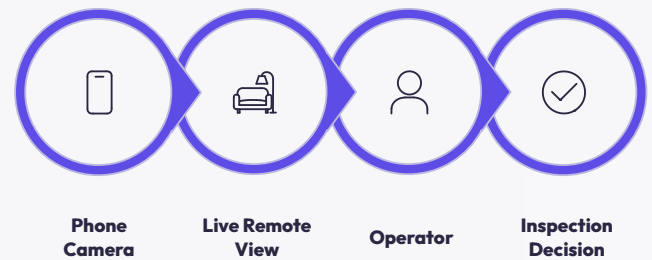
A REMOTE EYE INTO THE ENVIRONMENT.

The CyberRover X4.3 is equipped with a phone camera that provides live visual observation of the rover's surroundings. The operator views the camera feed in real time, enabling remote inspection of environments that are difficult or dangerous to access directly.

Current Visual Capabilities

- **Live Remote Camera Feed**
- **Visual Confirmation of Obstacles and Terrain**
- **Observation of Hazardous Locations**
- **Support for Navigation and Inspection Decisions**

Vision Pipeline



Future Development: A dedicated analog drone camera can replace or supplement the phone-camera arrangement in a future version, providing a more integrated onboard vision system with improved stability and field-of-view control.

THE OPERATOR

CONTROL WITHOUT DIRECT EXPOSURE.

The CyberRover X4.3 is controlled via a custom ESP32-based remote controller using the ESP-NOW wireless protocol. Commands flow to an Arduino Uno on the rover, keeping the operator safe while extending access into hazardous environments.



**ESP-NOW
Wireless Link**



**Dual-Axis
Joystick Input**



**OLED Cyber
OS Display**



**Real-Time
Telemetry**



Safety-Oriented Behavior

The remote controller operates silently with the buzzer disabled by design, and features dedicated **Back**, **Enter**, and **Save** button controls integrated into the Cyber OS interface.

DEPLOYMENT

WHERE CYBERROVER X4.3 CAN GO FIRST.



Industrial Inspection

Gas-leak detection, hazardous atmospheres, and restricted machinery areas.



Confined Spaces

Pipes, ducts, crawl spaces, tunnels, and narrow passages.



Mining & Hazardous Environments

Remote inspection in combustible gases, heat, and rough terrain.



Fire-Affected Areas

Post-fire assessment in damaged zones and nearby hazards.



Earthquake & Disaster Response

Inspect damaged structures and debris fields before responders enter.



Remote Reconnaissance

Observation of distant, uncertain, or difficult-to-access locations.



Primary civilian domains include search-and-rescue support and environmental hazard scouting. CyberRover X4.3 is designed for remote assessment in risky locations, not for military operations or battlefield deployment.

LIMITATIONS

EVERY ENGINEERING SYSTEM HAS BOUNDARIES.

The CyberRover X4.3 is a prototype — understanding its current limitations is essential for safe deployment and future development.



Environmental Sensing

MQ sensors provide raw analog indication only — not certified gas measurements.



Ultrasonic Sensing

Three forward sonar with blind spots, limited range, and no 3D LiDAR.



Communication

Dual-link system: Sub-GHz LoRa for telemetry, 2.4 GHz for motor control and video.



Autonomy & Intelligence

Fully operator-controlled with automated ultrasonic emergency stop — no SLAM or computer vision.



Temperature

DHT11 & BMP280 not rated for extreme heat environments.



Visual Inspection

Phone camera subject to field of view, lighting, and vibration constraints.



Mobility & Power

Ground clearance, chassis design, and 3S battery constraints limit operational reach.



Safety & Certification

University/exhibition platform — not explosion-proof or ATEX-rated.

ENGINEERING STATUS

ENGINEERED IN LAYERS. BUILT TO EVOLVE.

Built & Demonstrated (X4.3)

- 4WD mobility & dual BTS7960 drivers
- ESP32-S3 N8R16 Master Controller
- Arduino Uno low-level motor driver
- Arduino Nano environmental sensor deck
- MQ-4/MQ-7/MQ-135 multi-gas array
- DHT11 & BMP280 (Temp, Humidity, Pressure, Altitude)
- 3× ultrasonic collision radar sensors
- Phone-camera live visual feed
- 1.3" I2C OLED with procedural RoboEyes
- Reyax RYLR998 sub-GHz LoRa telemetry
- u-blox NEO-6M GPS positioning module
- ESP32 handheld remote with OLED Cyber OS
- Laptop ground cockpit with SQLite logging & CSV export

Integration & Refinement

- System reliability
- Sensor integration & noise isolation
- LoRa telemetry robustness
- Field testing over rubble
- Mechanical chassis refinement
- Power grid management & 3S voltage divider
- Wireless range verification
- Sensor threshold calibration
- Durability validation

Future Development

- Dedicated analog camera module
- Integrated onboard vision system
- Autonomous waypoint navigation
- Advanced multi-sensor obstacle avoidance
- Extended environmental sensor suite
- Modular mission payloads

MOVE. SENSE. INSPECT.

CYBERROVER X4.3

EXTENDING HUMAN AWARENESS INTO PLACES THAT ARE DIFFICULT, DANGEROUS OR
IMPOSSIBLE TO INSPECT DIRECTLY.
